

Temperature Controlled SWITCH

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This simple temperature controlled circuit triggers automatically when the ambient temperature goes beyond a set limit of, say, 50 degrees centigrade. This

temperature setting can be changed as per requirement through the potentiometer in the circuit.

The circuit has three parts: the sensor, main control unit, and relay

driver. It works on 230V AC mains. A bulb used in the circuit acts as a load for demonstration purpose. When the room temperature goes beyond the set limit the bulb switches on automatically. The author's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the temperature controlled switch is shown in Fig. 2. The circuit comprises step-down transformer X1, bridge rectifier BR1, 5V voltage regulator 7805 (IC1), temperature sensor LM35 (IC2), op-amp LM358 (IC3), 5V single changeover relay RL1, and a few other components. Capacitor C1

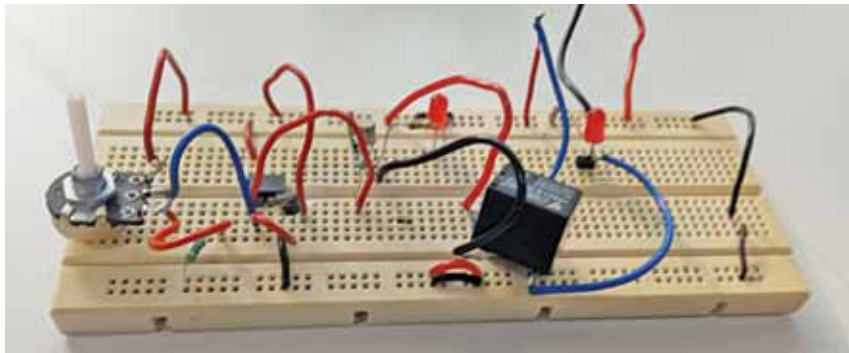


Fig. 1: Author's prototype on a breadboard

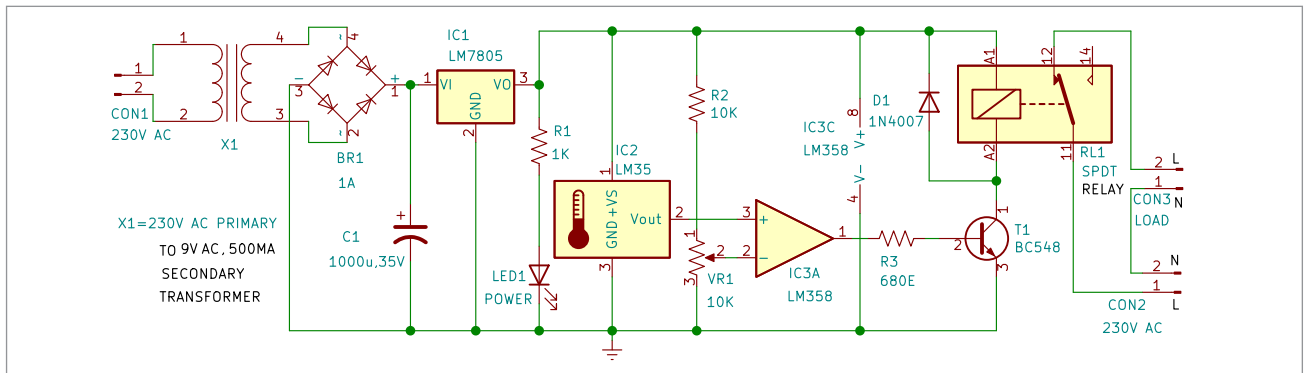


Fig. 2: Circuit diagram of the switch

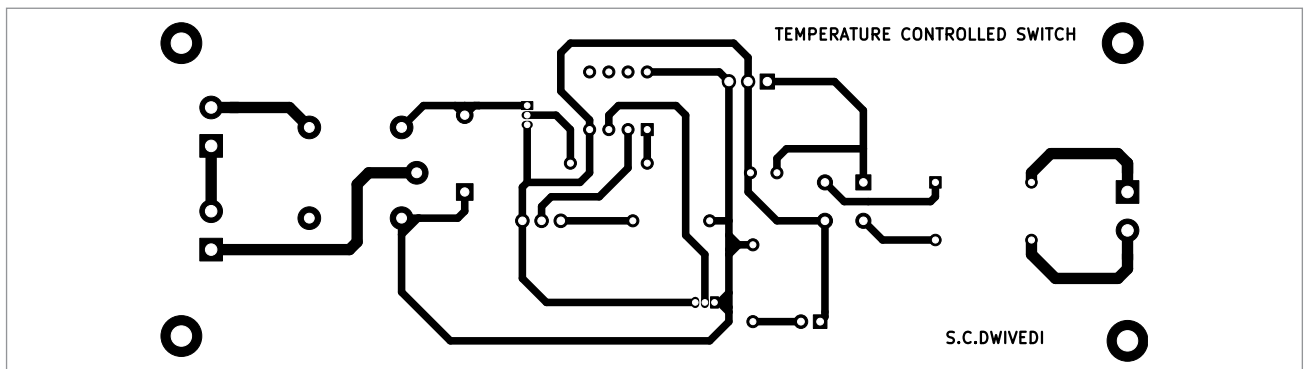


Fig. 3: Actual-size PCB layout for the circuit